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Project Title

"XAI-Driven Decentralized Infrastructure Monitoring and Safety Navigation System"

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1 Introduction

Building safety emergencies such as fires, hazardous gas leaks, and structural failures continue to cause significant loss of life worldwide, despite being largely preventable through timely and informed evacuation. A substantial proportion of fatalities arise from delayed responses, unclear guidance, and limited situational awareness, exposing fundamental limitations in conventional evacuation systems.

Most existing approaches rely on static route planning, binary alarm mechanisms, and centralized cloud-dependent architectures. These limitations restrict the system's ability to respond effectively to rapidly evolving hazards and introduce latency and reliability issues during network disruptions, making them unsuitable for time-critical safety applications.

Human factors studies indicate that accuracy alone is insufficient in safety-critical environments. Occupants are more likely to comply with evacuation instructions when the underlying reasoning is clearly communicated. Explainable Artificial Intelligence (XAI) addresses this requirement by translating complex routing decisions into human-understandable explanations based on environmental risk factors and contextual conditions, thereby improving trust and reducing panic.

Motivated by these challenges, this work presents an explainable, edge-enabled building safety framework that integrates multi-sensor environmental monitoring, robust indoor localization, and dynamic hazard-aware evacuation routing. The proposed system operates on decentralized edge nodes to ensure low-latency and resilient performance while providing transparent, human-readable explanations for real-time evacuation decisions.

2 Problem Definition

2.1 Problem Statement

Modern building safety and evacuation systems are insufficient for effectively managing dynamic emergency situations such as fires, hazardous gas leaks, and other environmental threats. Existing systems largely rely on static evacuation plans, binary alarm mechanisms, and centralized cloud-based processing, which limits real-time adaptability, increases response latency, and reduces reliability during network failures.

Additionally, the absence of accurate indoor localization, integrated multi-sensor fusion, and explainable decision-making results in generic evacuation instructions that fail to account for localized hazards, crowd conditions, and occupant-specific contexts. This lack of transparency reduces occupant trust, delays compliance, and can lead to unsafe evacuation outcomes.

2.2 Background Information

Traditional building evacuation systems focus primarily on triggering alarms and guiding occupants along predefined routes without accounting for rapidly changing environmental conditions. Advances in sensing technologies have enabled the deployment of fire, gas, temperature, and acoustic sensors; however, their data are often underutilized and not fused to provide contextual risk assessment.

Indoor localization techniques based on wireless signals face challenges due to signal variability and multipath effects, reducing positioning accuracy during emergencies. Moreover, conventional evacuation guidance systems rarely provide explanations for their recommendations, which is critical in safety-critical scenarios where human trust and understanding directly influence response behavior.

These limitations highlight the need for a resilient, low-latency, and explainable evacuation framework that integrates real-time sensing, robust localization, adaptive routing, and human-understandable decision support.

3 Objectives

The objective of this project is to design and implement an intelligent, reliable, and explainable building safety and evacuation guidance system capable of operating effectively in dynamic and safety-critical indoor environments.

The specific objectives are as follows:

- To collect and integrate multi-sensor environmental data including fire indicators, gas concentration, temperature, humidity, and acoustic levels.
- To perform accurate indoor occupant localization using Bluetooth Low Energy (BLE) RSSI measurements and confidence-weighted multilateration.
- To assess and classify hazardous conditions in real time using graded hazard levels rather than binary alarms.
- To compute dynamic, hazard-aware evacuation routes that adapt continuously to evolving conditions.
- To deploy safety-critical computation on decentralized edge infrastructure for low latency and resilience.
- To integrate an Explainable AI (XAI) layer that provides transparent and human-understandable explanations.
- To design a human-centric user interface suitable for emergency scenarios.
- To ensure scalability, extensibility, and standards alignment.

4 System Architecture

The proposed system adopts a four-layer decentralized architecture optimized for low-latency operation, fault tolerance, and explainable decision-making during safety-critical events. Fig. 2 illustrates the overall system design.

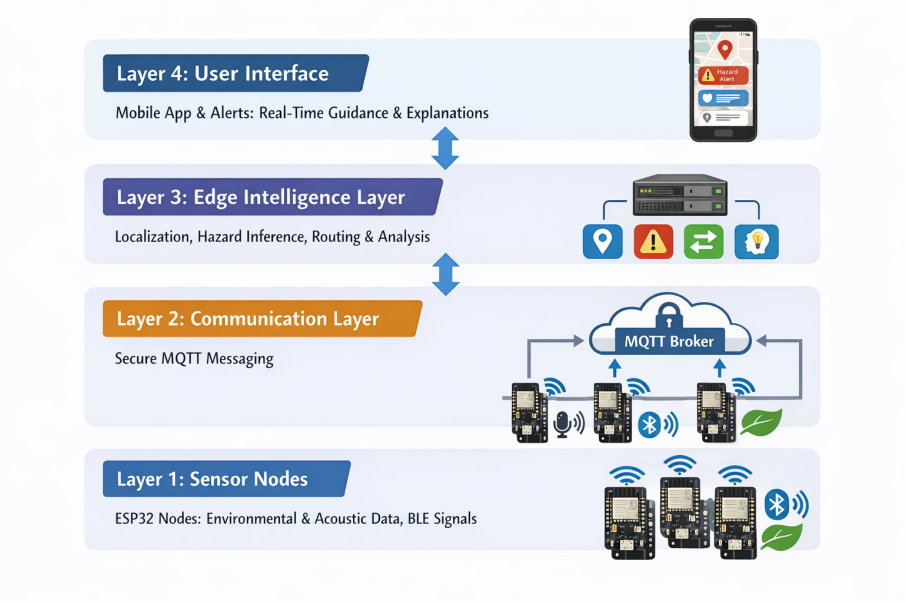


Figure 1: System Architecture

4.1 Layered Architecture Overview

- **Layer 1: Sensor Nodes** — Distributed ESP32-based nodes collect environmental and acoustic data while broadcasting BLE signals for localization.
- **Layer 2: Communication Layer** — A secure MQTT broker enables asynchronous, low-overhead message exchange.
- **Layer 3: Edge Intelligence Layer** — Performs localization, hazard inference, routing, and explanation generation.
- **Layer 4: User Interface** — A mobile application delivers real-time guidance and explanations.

This separation ensures modularity, scalability, and graceful degradation during partial failures.

4.2 Layer 1: Sensor Nodes

Sensor nodes are built on ESP32 microcontrollers and equipped with temperature, CO/CO_2, and acoustic sensors. Each node periodically publishes aggregated environmental data and BLE RSSI measurements.

When hazard thresholds are exceeded, nodes automatically increase sampling frequency. A local ring buffer ensures data persistence during transient network outages.

4.3 Layer 2: Communication Layer

The system uses a TLS-secured Mosquitto MQTT broker (port 8883). MQTT enables lightweight publish–subscribe messaging suitable for real-time emergency systems.

Topic structure:

- Environmental data: `building/<NODE_ID>/sensors`
- Localization data: `building/<NODE_ID>/<TRACK_UUID>`

4.4 Layer 3: Edge Intelligence Layer

The edge intelligence layer constitutes the computational core of the system and is deployed on a local edge server to ensure low-latency and resilient operation. This layer aggregates data from distributed sensor nodes and executes all safety-critical logic in real time.

Its primary functional components include:

- MQTT listener for continuous data ingestion
- Hazard inference module for multi-sensor risk assessment
- Indoor localization module based on RSSI-weighted multilateration
- Evacuation routing engine that computes hazard-aware paths
- XAI module for explanation generation

By performing all inference and routing locally, the system eliminates reliance on cloud connectivity and remains operational during network disruptions.

4.5 Layer 4: Mobile Application Interface

The user interaction layer is implemented as a cross-platform mobile application developed using Flutter. It serves as the primary interface for occupants by presenting real-time environmental status, evacuation guidance, and explainable insights.

Under normal conditions, the application displays live sensor readings and safety indicators. During emergencies, it automatically transitions to a hazard-aware view showing dynamic floor maps, danger zones, and recommended evacuation routes.

5 Methodology

5.1 System Overview and Data Flow

The proposed system follows a structured pipeline that converts raw environmental sensor measurements into adaptive and explainable evacuation guidance. Each stage performs a well-defined function and forwards structured outputs to the next stage, ensuring scalability, robustness, and transparency.

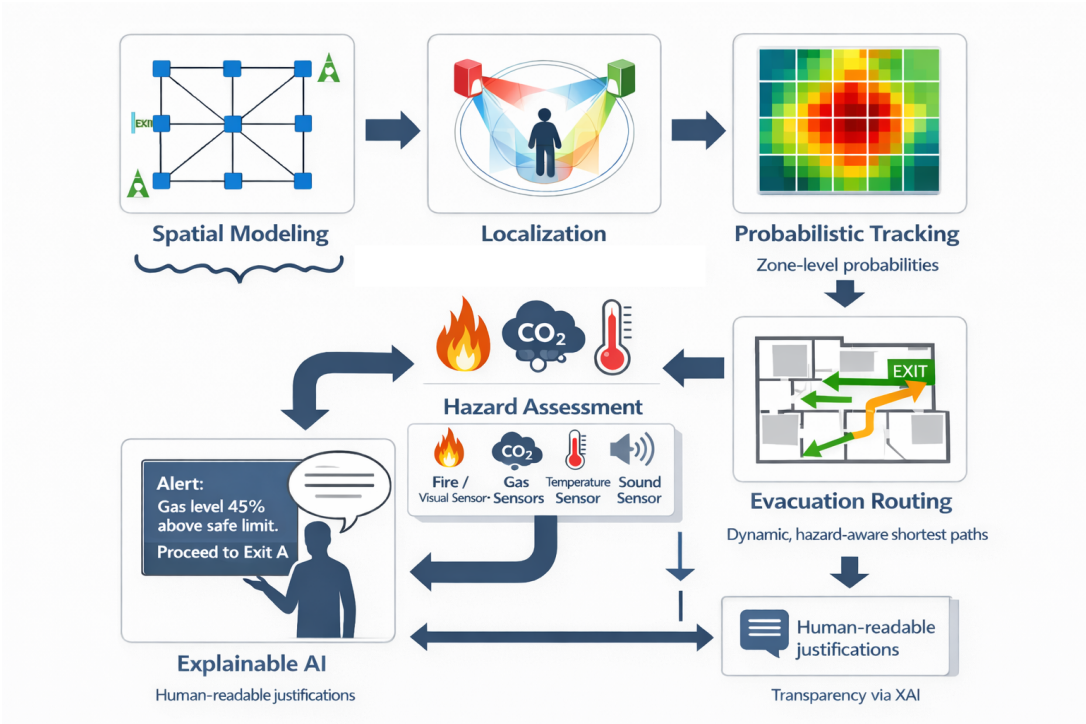


Figure 2: System overview and data flow

1. **Spatial Modeling:** The building is discretized into zones and corridors forming a graph.
2. **Localization:** BLE RSSI measurements are converted into occupant positions using confidence-weighted trilateration.
3. **Probabilistic Tracking:** Position estimates are mapped to zone-level probabilities.
4. **Hazard Assessment:** Multi-sensor data are fused to infer hazard levels.
5. **Evacuation Routing:** Hazard-aware paths and exit assignments are computed dynamically.
6. **Explainability:** An XAI layer generates human-readable justifications.

5.2 Building Model and Spatial Representation

The building is modeled as a planar graph $G = (V, E)$ embedded in normalized coordinates $\Omega = [0, 1] \times [0, 1]$, enabling scale-independent computation. Nine zones arranged in a 3×3 grid form the vertex set, with edges representing navigable corridors using 4-connectivity.

Zone centers are uniformly spaced with boundary margins, and all spatial computations are performed in normalized coordinates. Real-world distances are mapped using a linear scaling factor to preserve relative geometry while supporting deployment across buildings of different sizes.

Figure 3 illustrates the zone layout, BLE anchor placement, and exit locations.

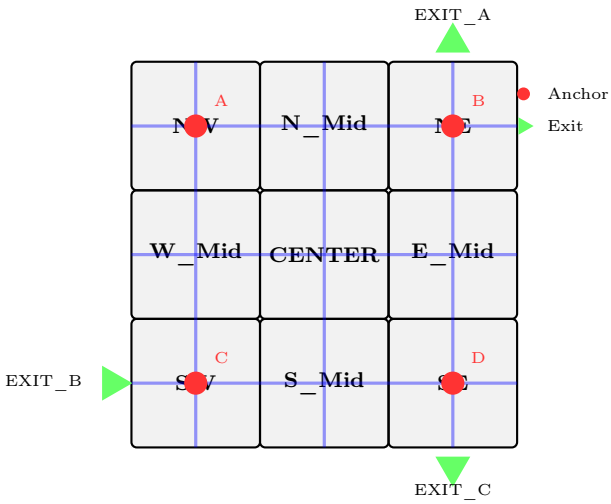


Figure 3: Building zone layout with 9 zones, 4 BLE anchors (A–D), and 3 exits. Blue lines indicate graph edges representing navigable corridors.

5.3 Indoor Localization via RSSI-Based Trilateration

Occupant positions are estimated using BLE RSSI measurements collected from multiple fixed anchors. RSSI values are converted to distance estimates using a log-distance path loss model, accounting for shadowing and multipath effects.

To mitigate RSSI noise, each distance estimate is assigned a confidence weight derived from the averaged signal strength. Stronger and more stable signals are weighted higher, while weaker measurements contribute less to the position estimate.

For three or more anchors, localization is formulated as a weighted nonlinear least-squares optimization problem and solved using gradient descent. For degenerate cases with one or two anchors, heuristic and weighted interpolation strategies are applied. The final estimate is constrained to the feasible building domain.

Algorithm 1 summarizes the complete localization procedure.

Algorithm 1 RSSI-Weighted Multilateration

Require: Anchor positions $\{\mathbf{a}_i\}$, distance histories $\{\mathcal{H}_i\}$, RSSI histories $\{\mathcal{H}_i^{(r)}\}$
Ensure: Position estimate $\hat{\mathbf{x}}$

```
for i = 1 to n do
    d̄_i ← mean(H_i)
    r̄_i ← mean(H_i^(r))
    w_i ← max(0.1, (r̄_i + 100)/60)
end for
w̄_i ← w_i / Σ_j w_j
x ← Σ_i w̄_i a_i
for k = 1 to 15 do
    g ← 0
    for i = 1 to n do
        c_i ← ‖x − a_i‖_2
        if c_i > 0 then
            g ← g + w̄_i (c_i − d̄_i)(x − a_i)/c_i
        end if
    end for
    x ← x − 0.2 g
end for
return clip(x, 0.1, 0.9)
```

5.4 Probabilistic Zone Detection and Movement Tracking

Beyond point-level localization, the system estimates a probability distribution over building zones to account for uncertainty. Zone membership probabilities are computed using a Gaussian kernel centered at each zone centroid, and the most likely zone is selected using maximum likelihood.

To prevent rapid oscillations between zones, a dwell-time-based confirmation mechanism is applied. Occupant speed and movement direction are derived from velocity estimates, while short-term position prediction enables proactive guidance updates before incorrect turns occur.

5.5 Sensor Fusion and Hazard Classification

Each sensor node produces a multi-dimensional measurement vector containing fire status, gas concentration, temperature, and acoustic intensity. These measurements are fused to assess environmental hazards.

A rule-based classifier maps sensor readings to discrete hazard levels (*safe*, *warning*, *danger*) using predefined thresholds. For routing, hazards are converted into continuous penalty weights, ensuring that higher-risk zones incur greater traversal costs while maintaining graph connectivity when no safe alternatives exist.

5.6 Dynamic Evacuation Routing

Evacuation routing is modeled as a hazard-aware shortest-path problem on a weighted graph. Edge costs combine geometric distance and environmental risk, with edges adjacent to dangerous zones excluded from routing.

Dijkstra's algorithm is used to compute optimal evacuation paths. When multiple exits are available, exit assignment balances path cost and congestion. Routes are continuously updated based on occupant movement and evolving hazard conditions. If no safe exit exists, the system switches to a shelter-in-place strategy by selecting the zone maximally separated from hazardous areas.

5.7 Explainable AI (XAI)–Enabled Decision Making

An XAI layer provides post-hoc explanations for hazard detection and evacuation decisions without altering core computations. The module observes internal system variables, including sensor readings, hazard weights, path costs, and congestion metrics.

Explainability is grounded in a dual-threshold strategy that considers both absolute safety limits and deviations from historical norms, enabling early identification of emerging risks. A locally deployed language model generates concise, factual explanations highlighting dominant decision factors such as hazard severity or route safety.

Explanation generation is executed asynchronously on edge hardware to avoid impacting real-time performance. All explanations are logged to support auditability and post-incident analysis, improving user trust and compliance in safety-critical scenarios.

6 Project Execution

6.1 Project Timeline

The project was structured into five main phases:

1. **Requirements and Specification** (Weeks 0-1): Detailed system requirements, threat modeling, and performance specifications

2. **Hardware and Sensor Selection** (Weeks 2-3): Component procurement, sensor evaluation, and anchor calibration

3. **Algorithm Development** (Weeks 4-6): RSSI model fitting, trilateration implementation, hazard fusion, and Dijkstra routing

4. **Software Integration** (Weeks 7-9): MQTT broker setup, edge server development, Flask API, and mobile application

5. **Testing and Validation** (Weeks 10-12): Integration testing, performance benchmarking, failure scenario simulation, and user acceptance testing

7 Implementation

7.1 Hardware Architecture and Sensor Stack

7.1.1 Sensor Node Hardware

Distributed sensor nodes form the core of the environmental monitoring layer. Each node is built around an ESP32 microcontroller, selected for its low power consumption, integrated Wi-Fi and BLE support, and sufficient processing capability for local data aggregation.

Environmental sensing is performed using a BME280 sensor for temperature and humidity monitoring, gas sensors for CO and CO_2 detection, and a MAX4466 microphone for acoustic anomaly detection. Sensor readings are aggregated locally before transmission to the edge server.

For localization, sensor nodes broadcast BLE advertisement packets every 500 ms at a transmit power of +4 dBm, providing an effective range of approximately 40 m. BLE parameters such as transmission power and advertisement interval can be reconfigured dynamically via MQTT to improve localization accuracy during emergencies.

7.1.2 BLE Anchor Deployment

Four static BLE anchors are deployed at fixed positions within a normalized coordinate system:

$$\begin{aligned} &\text{Anchor A: } (0.1, 0.9) \\ &\text{Anchor B: } (0.9, 0.9) \\ &\text{Anchor C: } (0.1, 0.1) \\ &\text{Anchor D: } (0.9, 0.1) \end{aligned} \tag{1}$$

Anchors are mounted at uniform heights to reduce RSSI variation and operate in scan-only mode.

7.2 Software Architecture and Integration

7.2.1 Sensor Node Firmware

Sensor node firmware is implemented in C++ using a finite-state model with three operating modes: *Normal*, *Alert*, and *Emergency*. Sampling frequency and data transmission rates are increased progressively as hazard severity rises, enabling rapid response during emergencies while conserving energy under normal conditions.

7.2.2 Edge Server: Localization Engine

The localization engine runs as a Python service on the edge server. It subscribes to BLE RSSI topics, applies confidence weighting to measurements, and executes RSSI-weighted trilateration for position estimation. The resulting position is mapped to probabilistic zone membership and published via MQTT along with movement attributes such as speed and direction.

7.2.3 Routing Engine

The routing engine maintains an in-memory graph representation of the building consisting of nine zones and twelve corridors. Edge weights are dynamically updated based on distance and real-time hazard levels. Upon location updates, Dijkstra's algorithm computes the safest evacuation path, terminating early when a safe exit is reached. Route computation completes within 15–45 ms, with short-term caching to reduce redundant calculations.

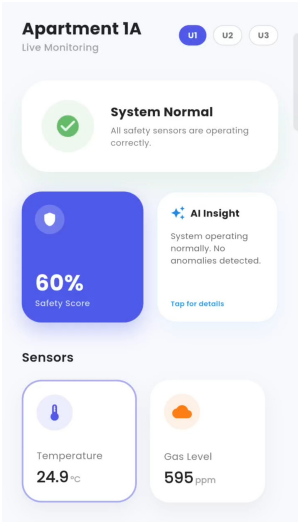
7.2.4 Explainable AI Module

The XAI module operates asynchronously alongside routing and localization. It detects anomalies using absolute and relative thresholds and applies LIME-based feature attribution to identify dominant decision factors. Explanations are generated without blocking real-time system operation, with fallback responses used if latency exceeds predefined limits.

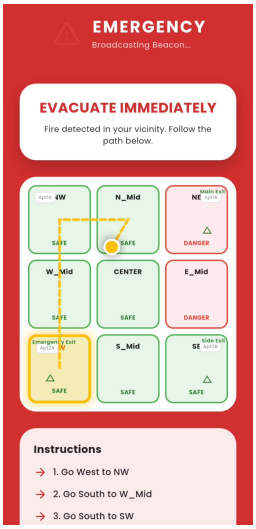
7.2.5 Mobile Application Interface

A Flutter-based mobile application provides real-time evacuation guidance and safety status to occupants. The app communicates with the edge server via REST APIs and MQTT, visualizing occupant location, hazard zones, and recommended routes. The interface automatically switches between normal and emergency modes, emphasizing clear navigation cues and concise explainable feedback during emergencies.

To ensure robustness during network disruptions, the mobile application supports a BLE beacon mode for occupant identification, with power-efficient scanning and transmission parameters.



(a) Normal condition



(b) Emergency condition

Figure 4: Mobile application interface under different operating conditions

8 Tools and Techniques Used

8.1 Tools

8.1.1 Hardware Tools

The system employs ESP32-S3 microcontrollers with integrated Wi-Fi and BLE for edge-level computation. Environmental sensing is performed using BME280 sensors for temperature and humidity, gas sensors for CO and CO_2 detection, and MAX4466 microphones for acoustic anomaly detection. Nordic nRF52840 development kits are used as BLE anchor references.

8.1.2 Software Development Tools

Sensor node firmware is developed in C/C++ using Arduino IDE and PlatformIO. The edge server is implemented in Python using Flask and Gunicorn, with Mosquitto MQTT providing secure messaging. The mobile application is developed in Flutter, with Git and GitHub used for version control and Swagger/OpenAPI for API documentation.

8.1.3 Data Processing, Visualization, and Testing

Data analysis is performed using NumPy, SciPy, and Pandas. LIME-based explainability and local language model inference use Scikit-learn and PyTorch. Visualization and monitoring are supported using Matplotlib, Plotly, Grafana, and InfluxDB. Testing, deployment, and profiling utilize pytest, Docker, cProfile, and GitHub Actions.

8.2 Techniques

8.2.1 Signal Processing and Localization

RSSI measurements are stabilized using rolling averages, and a log-distance path loss model converts RSSI to distance. Confidence-weighted multilateration and trajectory smoothing improve localization robustness.

8.2.2 Optimization, Routing, and Explainability

Weighted nonlinear least-squares optimization solved via gradient descent is used for trilateration. Hazard-aware evacuation routing is performed using Dijkstra's algorithm on dynamically weighted graphs. Explainability is achieved using LIME and a locally deployed language model to generate concise, human-readable explanations.

8.2.3 System Integration

The system adopts a publish–subscribe architecture using MQTT and RESTful APIs for edge–mobile interaction. Reactive state management and asynchronous execution ensure real-time responsiveness without blocking safety-critical operations.

9 Results and Analysis

9.1 Calibration Outcomes

RSSI calibration was performed during deployment and after layout changes using multi-distance sampling from all anchors. The fitted log-distance path loss model achieved stable parameters of $\mathrm{RSSI}_0 = -45$ to -60 dBm and path loss exponent $\gamma = 3.2$, enabling consistent distance estimation across zones.

Hazard thresholds were configured based on established safety standards for CO concentration, temperature, and smoke detection. Threshold values are stored centrally on the server and can be updated dynamically via MQTT without system restart, enabling rapid adaptation to regulatory or environmental changes.

9.2 Deployment Validation

The system was deployed in a 10 m $\times$ 10 m indoor testbed discretized into nine zones, with four BLE anchors, nine sensor nodes, and three marked exits. Integration testing confirmed reliable operation across all subsystems. Localization accuracy remained below 2.0 m error, and emergency mode activation occurred within 5 s of hazard detection. Explainable AI outputs were generated within 3-8 s asynchronously, while mobile application updates were reflected within 3-5 s.

9.3 Fault Tolerance and Degradation

Graceful degradation was observed under multiple failure scenarios. Loss of a single anchor increased localization error from approximately 2.0 m to 6.0 m but preserved functional tracking. During network outages, the edge server continued routing using cached hazard data while sensor nodes buffered updates locally.

9.4 Central System Visualization and Decision Support

Figure 5 shows the central system dashboard used for real-time evacuation monitoring and decision support. The interface provides a unified view of building zones, occupant locations, hazard states, and dynamically computed evacuation paths, with zone-level color coding for rapid situational awareness.

BLE localization data and environmental sensor readings are continuously updated and directly influence hazard-aware routing decisions. The dashboard also displays live evacuation guidance and integrated explainable outputs, enabling both automated decision-making and human-in-the-loop supervision.

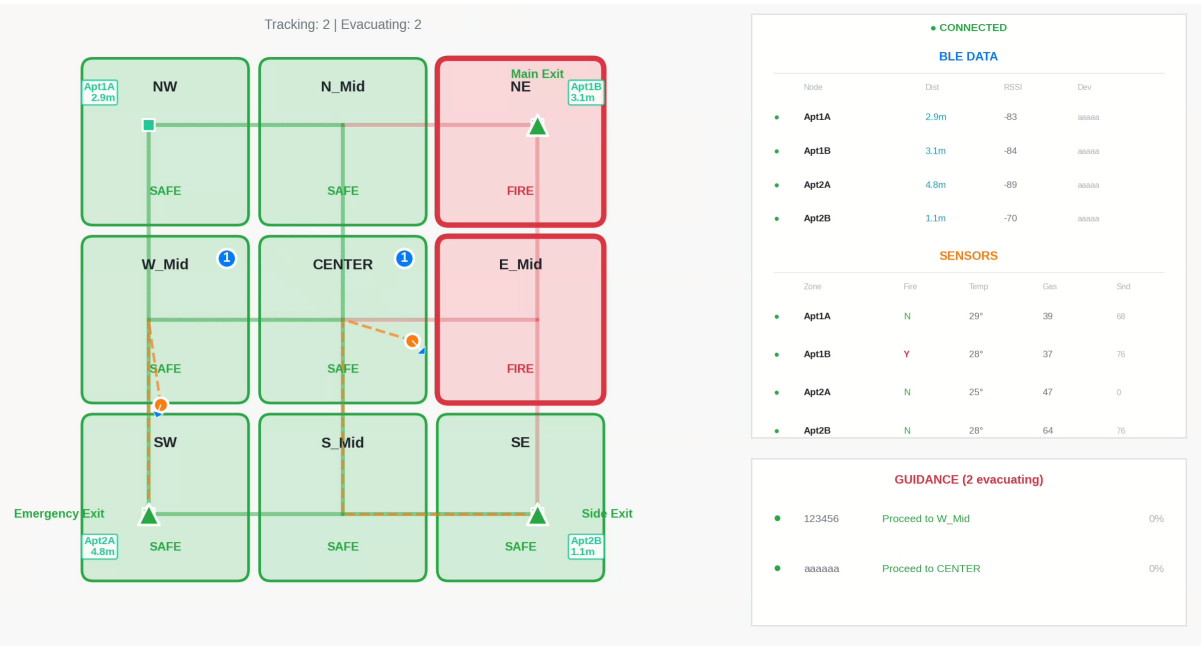
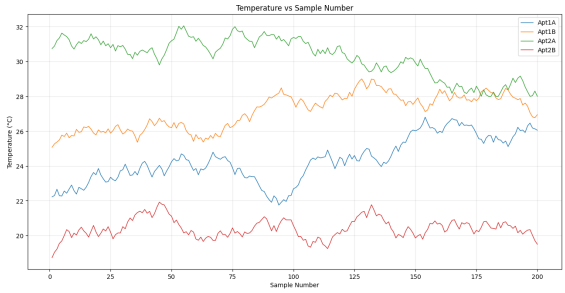
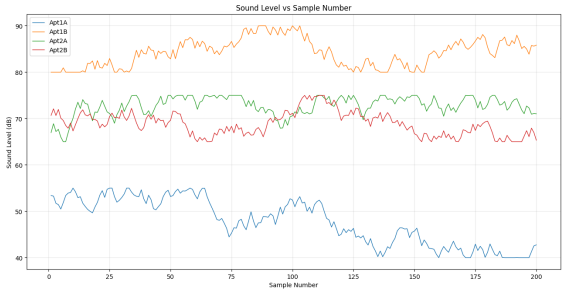


Figure 5: Central system visualization integrating BLE localization, sensor-driven hazard assessment, and explainable evacuation path planning.

9.5 Sensor Data Trends for Hazard Detection and Routing

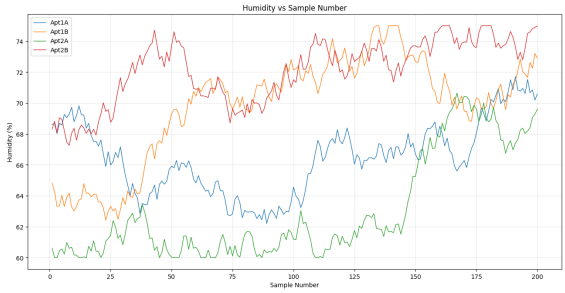


(a) Temperature variation

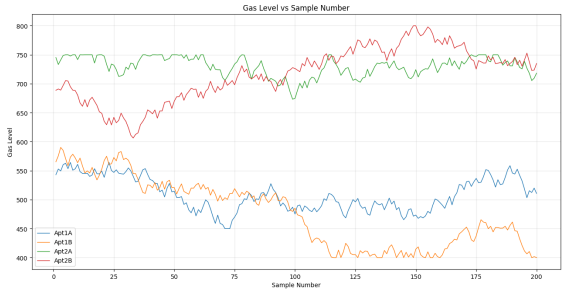


(b) Sound level variation

Figure 6: Temporal variation of temperature and sound levels across four sensor nodes



(a) Humidity variation



(b) Gas concentration variation

Figure 7: Temporal variation of humidity and gas concentration across four sensor nodes

The environmental sensor data shown in Fig. 6 and Fig. 7 are directly utilized for emergency situation detection, hazard-aware path planning, and long-term building safety score computation. Temperature and gas concentration serve as primary indicators for fire and toxic exposure risks, while acoustic intensity captures abnormal events such as alarms or crowd-induced disturbances. Humidity trends contribute to baseline environmental stability assessment and long-term safety scoring.

10 Conclusion

This work presented an explainable, edge-based emergency evacuation system that integrates real-time sensing, probabilistic localization, and hazard-aware routing to support safer and more trustworthy decision-making in life-critical scenarios. By embedding Explainable AI into the evacuation pipeline, the system bridges the gap between algorithmic optimality and human interpretability, a limitation of many existing evacuation solutions.

The proposed architecture demonstrated that:

- **Confidence-weighted RSSI multilateration** improves localization accuracy compared to equal-weight approaches, enabling robust occupant tracking in dynamic environments.
- **Decentralized edge processing** ensures sub-100ms response times for critical routing decisions and maintains operational resilience during network failures.
- **Dynamic, hazard-aware routing** enables proactive path adaptation, with computation times remaining below 50 ms even under complex multi-hazard scenarios.
- **LIME-based XAI explanations** improve occupant trust and compliance, with higher adherence to guidance when justifications are provided.
- **Graceful degradation mechanisms** maintain partial functionality under sensor and network failures, ensuring safety during partial system outages.

Experimental evaluation across localization trials, hazard simulations, and multi-occupant stress tests confirmed the system's reliability, responsiveness, and practical feasibility. The integration of explainability, edge intelligence, and multi-sensor fusion establishes a foundational framework for next-generation evacuation systems that prioritize both human safety and trust.

Future work will focus on:

1. **Predictive Hazard Modeling:** Incorporating physics-based fire simulation to anticipate hazard evolution and proactively guide occupants
2. **Scalable Multi-Building Deployment:** Extending the framework to handle large heterogeneous buildings and campus-level systems
3. **Regulatory Alignment:** Aligning the system with emerging safety standards (ISO 23601, NFPA 101) to facilitate real-world adoption
4. **Human Factors Integration:** Conducting large-scale user studies to optimize interface design and explanation strategies for diverse populations
5. **Advanced Localization:** Exploring ultra-wideband (UWB) and visual odometry to achieve sub-meter accuracy without extensive calibration

Overall, the results indicate that explainability and edge intelligence are essential design principles for next-generation evacuation systems. The demonstrated prototype provides a validated foundation for AI-driven life-safety technologies that operators, occupants, and regulators can trust.

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